

SHRUTI VANPARIA

AI/ML Engineer

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PROFESSIONAL SUMMARY

Generative AI and AI/ML Engineer with 3+ years of experience developing and supporting enterprise AI applications, with a focus on LLM applications, RAG pipelines, agentic AI workflows, conversational AI, machine learning APIs, and production AI solutions. Hands-on experience with Python, FastAPI, LangChain, LangGraph, LlamaIndex, Azure OpenAI, Gemini, Amazon Bedrock, Dialogflow CX, vector databases, embeddings, and REST APIs. Experience in building retrieval and search solutions, integrating LLMs with enterprise applications, developing AI inference services, and implementing prompt engineering, tool calling, structured outputs, and evaluation frameworks. Skilled in RAGAS, DeepEval, MLflow, Docker, GitHub Actions, AWS, Google Cloud, Pytest, SQL, and model monitoring. Experienced in troubleshooting AI application issues, evaluating model and retrieval performance, improving system reliability, and collaborating with engineering, data science, QA, and product teams to deploy scalable AI solutions in production environments.

TECHNICAL SKILLS

Generative & Agentic AI: LLMs, RAG, Agentic AI, Prompt Engineering, Embeddings, Function/Tool Calling, Structured Outputs, MCP, LangChain, LangGraph, LlamaIndex, OpenAI, Azure OpenAI, Anthropic, Gemini, Amazon Bedrock, Dialogflow CX

Vector Search & Retrieval: pgvector, FAISS, Hybrid Search, Semantic Search, Reranking, BERT Embeddings

Programming & Backend: Python, SQL, Java (Spring Boot), FastAPI, REST APIs, Pydantic, Pytest, Streamlit, React

Machine Learning & NLP: scikit-learn, XGBoost, SHAP, PySpark, Classification, Regression, Forecasting, NLP, TF-IDF

Cloud & MLOps: AWS (SageMaker, ECS, S3), Databricks, Docker, Git, GitHub Actions, Azure DevOps, CI/CD, MLflow, Langfuse, OpenTelemetry, Model Monitoring

LLM Evaluation & AI Safety: RAGAS, DeepEval, promptfoo, LLM-as-a-Judge, Golden Datasets, Hallucination Evaluation, Prompt-Injection Defense, Guardrails

Data & Analytics: PostgreSQL, MySQL, SQL Server, pandas, NumPy, ETL/ELT, Data Modeling, Data Validation, R, Power BI, Tableau, Excel

Open-Weight Models: Llama, Mistral, Qwen, Gemma, Ollama, vLLM, GGUF, AWQ, LoRA, QLoRA

PROFESSIONAL EXPERIENCE

Verizon • AI/ML Engineer Dallas, TX

May 2024 – Present

- Designed and deployed enterprise Conversational AI and Generative AI solutions using Dialogflow CX/ES, Azure OpenAI, and Retrieval-Augmented Generation (RAG) to support billing, technical support, and account-servicing workflows.
- Developed Dialogflow CX flows using intents, entities, routes, session parameters, playbooks, and fallback handling to improve customer self-service and intent recognition.
- Built FastAPI and XGBoost inference endpoints for real-time fraud detection, applied SHAP for model explainability, and worked with data scientists on feature refinement that contributed to a measurable reduction in fraud false positives.
- Integrated BERT-based embedding models into product search and ranking APIs and evaluated retrieval performance to improve search relevance.
- Contributed to RAG workflows covering embeddings, retrieval, prompt engineering, and context-grounded responses.
- Developed REST APIs that expose AI/ML capabilities to downstream applications; wrote Pytest unit and integration tests and maintained API documentation.

- Processed training datasets with PySpark on Databricks and supported ML deployments on AWS (SageMaker, ECS, S3).
- Contributed to MLOps workflows using MLflow for model tracking, Docker for containerization, and GitHub Actions for CI/CD; supported deployment, monitoring, and rollback across staging and production.
- Diagnosed production issues involving inference latency, API performance, and service reliability while collaborating with product, engineering, QA, and data science teams in Agile sprints.

Shreeji Plast Pvt. Ltd • Software Engineer

Gujarat, India Apr 2023 – Jul 2024

- Developed backend REST APIs using Java Spring Boot for telecom and BFSI clients, supporting billing, account management, and service-request workflows.
- Implemented authentication, input validation, structured error handling, and business validations; used SQL Server for data persistence and querying.
- Integrated ML-based intent classification (Logistic Regression, Random Forest) into chatbot backend services.
- Applied TF-IDF feature engineering and preprocessing improvements that increased F1-score from 0.74 to 0.82 for automated query classification.
- Refined training data and preprocessing workflows, helping reduce chatbot fallback rates by 10–12%; built scripts to convert raw chatbot logs into cleaned datasets for evaluation and retraining.
- Evaluated models using precision, recall, and confusion-matrix analysis, and collaborated with QA to validate outputs against intent categories and business requirements.
- Containerized backend services with Docker, supported CI/CD deployments through Azure DevOps, and resolved production issues involving API timeouts and application stability.

AI ENGINEERING PROJECTS

RecallRadar – Human-in-the-Loop Product Recall Agent Python, Streamlit, Strands Agents SDK, Gemini, Amazon Bedrock, CPSC API, Pytest, Docker- Built an agentic workflow that checks household products against the official U.S. CPSC recall feed and verifies exact model, lot, and UPC evidence. Designed a four-stage workflow (observe, investigate, verify, escalate) with weak-match suppression and approve/dismiss gates to keep safety decisions auditable. Added CSV/JSON decision exports, deterministic fallback logic, and automated tests for matching and safety boundaries.

GhostFrame AI – Agentic Media Provenance Verifier Python, Fast API, Vertex AI Gemini, Parallel Search API, Pydantic, HTTPX, Docker- Built a FastAPI service that investigates whether a movie trailer presented as official is supported by public evidence, using Gemini for investigation planning and Parallel Search to retrieve supporting and contradictory sources. Designed a fallback workflow using YouTube publisher metadata, official/unofficial channel signals, title-language cues, and reference context. Maintained explainable results when external AI or search services were unavailable.

ParSafe – Retrieval-Grounded Parking Decision Assistant Python, Fast API, React, Gemini, Docker, Pytest, GitHub Actions- Built a React and FastAPI parking assistant using Python ranking and retrieval logic over a processed parking dataset. Added an optional Gemini layer for retrieval-grounded responses while retaining a deterministic non-LLM processing path. Developed API documentation, automated tests, a Dockerized backend, and a static GitHub Pages demonstration.

Dental Operations Strategist – No-Show and Revenue What-If Planner Python, scikit-learn, Fast API, Streamlit- Built a FastAPI and Streamlit application serving no-show predictions and what-if scenarios for clinic scheduling and follow-up planning. Combined model scores with capacity, scheduling, and revenue rules to provide explainable operational outputs. Separated training and inference workflows and implemented data and model validation.

EDUCATION

Master of Science in Business Analytics and Artificial Intelligence

The University of Texas at Dallas, Naveen Jindal School of Management | Richardson, TX | May 2026 | GPA: 3.8/4.0

Bachelor of Science in Data Science

Indus University | Ahmedabad, India | May 2024 | GPA: 3.8/4.0